

Welcome

AI for Global Health

Introduction

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

EPFL

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BiGHT - Building Intelligent Global Health and Humanitarian response technologies

Introduction

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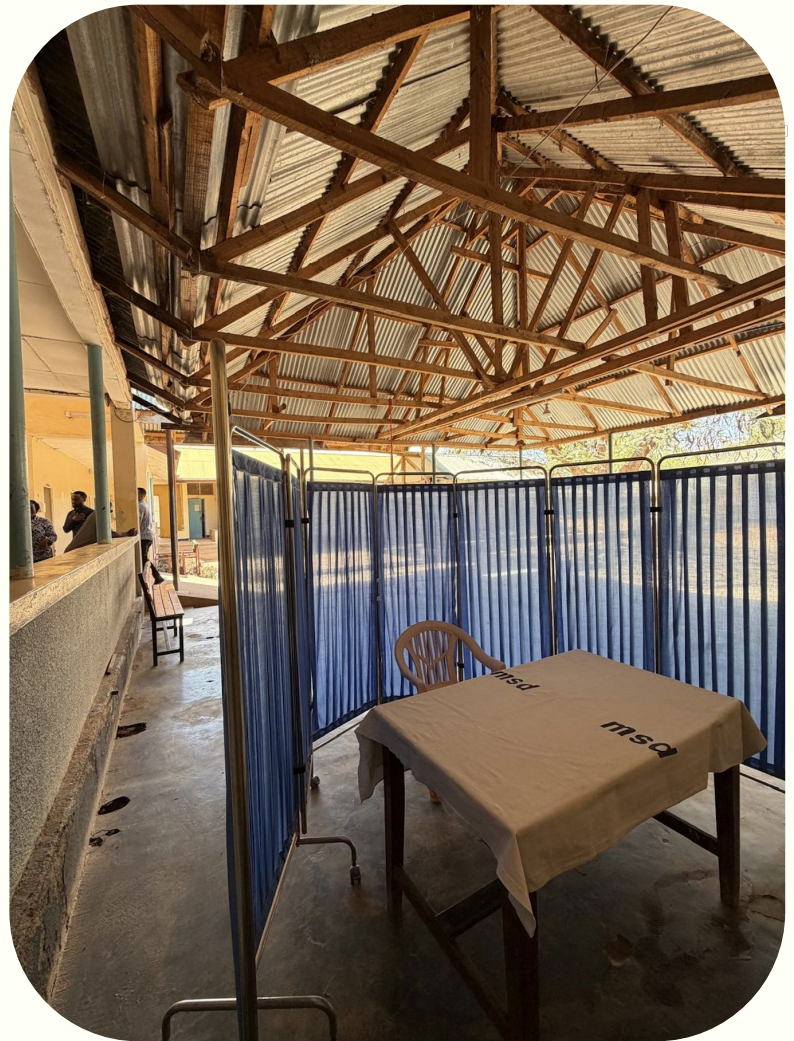
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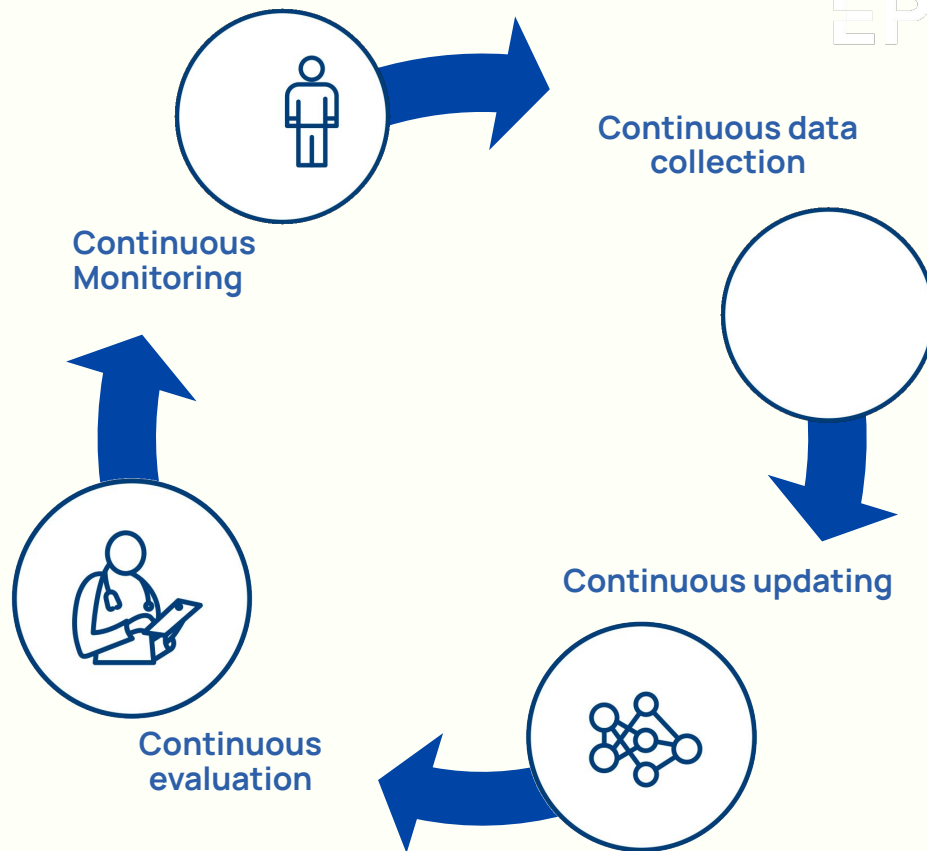




Welcome

EPFL

How we work



Welcome

EPFL

Where we focus



Welcome

What we make

Our work combines strong engineering, evaluation, and implementation science to build AI that works in the real world.

(meditronFO, MOOVE, MoBayes, DISCO, AntibioGo)

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.



Welcome

Who we are



Annie (Mary-Anne) Hartley

Director



Kristina Keitel

Clinical Director



Lars Klein

Academic Director



David Sasu

Technology Director

Contact information

Fabrice: fabrice.nemo@epfl.ch
Annie: mary-anne.hartley@epfl.ch



Fabrice Nemo

Laboratory for intelligent
Global Health & Humanitarian Academic Coordinator
Response Technologies.



Yusuf Kesmen

EPFL AI Center PhD Fellow



Jeremy Baffou

External TA



Xavier

Theimer-Lienhard

PhD Candidate

LiGHT

Paired lectures

Block	Weeks	Goal
Volatile Contexts	1-4	Understand the environments in which AI must operate, and why technology often fails in humanitarian and clinical settings.
High-stakes Decisions	5-7	Understand how clinicians make decisions under uncertainty and how AI can safely support human expertise.
Trustworthy Evidence	9-13	Design, evaluate, and deploy AI systems that are safe, effective, and ready for real-world use.

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
1	Sep 9	Inequitable Inaccuracy	Tutti Fratelli: The Principles That Created a Movement	Building for Broken Environments
2	Sep 16	When the Hospital Is the Target	When the Hospital Is the Target	Systems That Work When Nothing Else Does
3	Sep 23	The Last Mile	Health for All, Care for Few	The Last Mile: Getting Intelligence to the Edge
4	Sep 30	Listening Before the Outbreak	The Map That Stopped an Epidemic	Listening to the World: Epidemic Intelligence at Scale

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
5	Oct 7	Models Without Patients	The Machine That Was Right and Was Never Used	From Rules to Reasoning
6	Oct 14	Reasoning Against Ourselves	The Confident Fool: How Expertise Breeds Error	Debiasing the Machine
7	Oct 28	Advice at the Wrong Time	Why Doctors Ignore Good Advice	Integration Without Interruption

Paired lectures

Week	Date	Theme	Context lecture	Engineering lecture
9	Nov 11	The Fragility of Accuracy	How We Learned to Count What We Were Losing	When 99% Accuracy Is Dangerous
10	Nov 18	Causal Evidence in Practice	The Man Who Asked Medicine to Prove Itself	Causal Inference: Beyond Correlation
11	Nov 25	Designing for Failure	The Holes in the Cheese	Uncertainty as Architecture
12	Dec 2	Responsibility in Clinical AI	First, Do No Harm - Then, Prove It	Governing What You Cannot Fully Understand
13	Dec 9	Systems That Reach People	Why Good Ideas Die at the Door	Shipping to Scale: The MLOps of Global Health AI

Course Logistics



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Communication during the semester

The screenshot shows the Moodle interface for the course 'AI for global health / BiGHT'. The top navigation bar includes the EPFL logo and links for 'Accueil', 'Tableau de bord', 'Mes cours', and 'My Media'. The course title is displayed in red, followed by a breadcrumb trail 'Informatique (IN) / IN - Master'. Below the title, a horizontal menu contains 'Cours', 'Paramètres', 'Participants', 'Notes', 'Rapports', and 'Plus'. The main content area is divided into two sections: 'Généralités' and 'Discussion'. The 'Généralités' section includes links to the 'Course website', 'Anonymous Feedback Form', and 'Project application' (with a deadline of September 16, 2026), and a note about adding a GitHub handle. The 'Discussion' section includes a link to 'Ed Discussion'.

EPFL Accueil Tableau de bord Mes cours My Media

Informatique (IN) / IN - Master

AI for global health / BiGHT

Cours Paramètres Participants Notes Rapports Plus

Généralités Tout replier

- [Course website](#)
- [Anonymous Feedback Form](#)
- [Project application](#)
S'ouvre : mercredi 9 septembre 2026, 15:00 Se termine : mercredi 16 septembre 2026, 22:00
- Please tell us your GitHub handle to add you to the right repository

Discussion

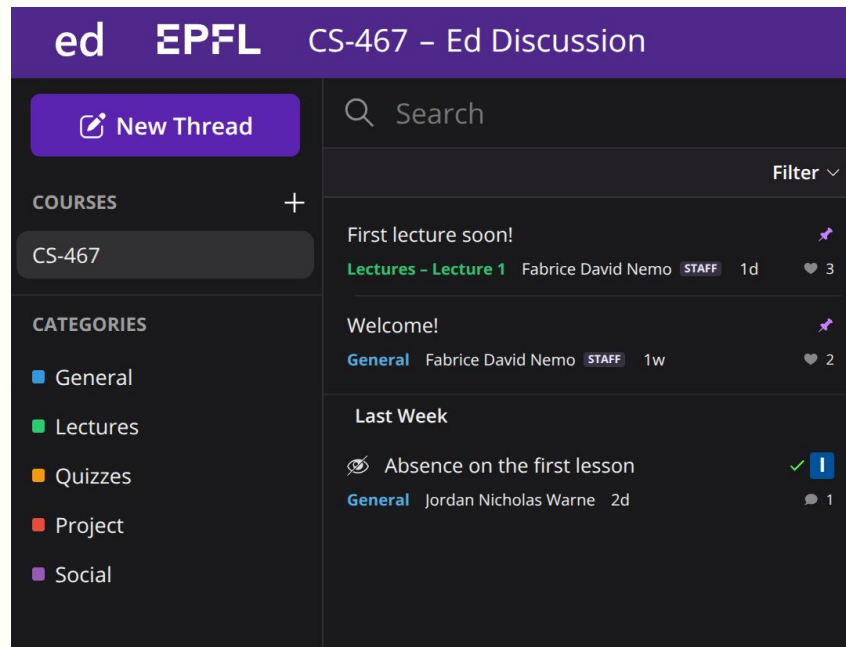
- [Ed Discussion](#)

Communication during the semester

Let's use the ed discussion forum for communication

Use Categories to sort your questions

Check whether your question has already been posted



Website

Course overview

PROJECT
Project Documentation
Project Briefs
Available after Sep 9, 15:00

Anonymous Feedback

BLOCK I - VOLATILE CONTEXTS
Week 1 - Inequitable Inaccuracy
Available after Sep 9, 15:00
Week 2 - When the Hospital Is the Target
Available after Sep 16, 15:00
Week 3 - The Last Mile
Available after Sep 23, 15:00
Week 4 - Listening Before the Outbreak
Available after Sep 30, 15:00

BLOCK II - HIGH-STAKES DECISIONS
Week 5 - Models Without Patients
Available after Oct 7, 15:00
Week 6 - Reasoning Against Ourselves
Available after Oct 14, 15:00
Week 7 - Advice at the Wrong Time
Available after Oct 28, 15:00
Week 8 - Midterm
Available after Nov 4, 15:00

BLOCK III - TRUSTWORTHY EVIDENCE
Week 9 - Fragility of Accuracy
Available after Nov 11, 15:00



Building intelligent Global Health and Humanitarian Response Technologies (BiGHT)



Step 1 of 5

Previous

1. Summary

Next (Course Overview)

Summary

CS-467 Course Curriculum - Fall 2026 - EPFL IC

Summary

This course teaches the principles and practice of designing, building, and rigorously evaluating trustworthy, representative, and contextually appropriate AI systems for high-stakes decision making in global health and humanitarian response.

Lecturers: Prof. Mary-Anne Hartley, Prof. Kristina Keitel, Dr. Lars Klein, Dr. David Sasu, Tim Arni, Yusuf Kesmen, Fabrice Nemo

Coordinator: Fabrice Nemo

TAs: Yusuf Kesmen, Jérémy Baffou

Website

Week 1: Inequitable Inaccuracy

1. Overview
1. Overview
2. Learning Objectives
3. Core Ideas and Case Anchor
4. Why Inequity Produces Inaccuracy
5. Humanitarian Principles as Design Constraints

Step 1 of 10

Previous

1. Overview

Next (Learning Objectives)

Overview

Block I: Volatile Contexts (Weeks 1-4)

Central question: Why do systems fail the people who need them most?

Big idea: Inequity is not background noise. It decides who appears in our data, who reaches care, and who is harmed when we optimize for the people who are already visible.

Block focus: Understand the environments in which AI must operate, and why technology often fails in humanitarian and clinical settings.

Giving Feedback



Anonymous Feedback 

BiGHT Feedback Form

This form lets you send the course team anonymous feedback, so that the course can be improved even before the official EPFL indicative feedback form.

fabrice.nemo@epfl.ch [Changer de compte](#)



 Non partagé

Overall, how satisfied are you with the course so far?

I hate it 1 2 3 4 5 I love it

☐ ☐ ☐ ☐ ☐

How useful have the lectures been for helping you understand the stakes in Global Health / Humanitarian Response and the engineering means for tackling them?

Not useful, I barely learnt anything and had to search for other sources 1 2 3 4 5 The lectures were quite informative and I could effectively learn from them

☐ ☐ ☐ ☐ ☐

What should we keep, change, add, or remove from the lectures?

Real world project

Please stay for the exercise session for a presentation of each project, you will apply and make groups by the end of next week

Project number	Project title		Team size	Short description
Project 1	Offline Speech-To-Speech System For Humanitarian Triage		5	Offline speech-to-speech system that can help humanitarian workers in Yemen with triage.
Project 2	Interactive Public Health Messenger		5	Build an interactive public health messenger that helps people understand crisis instructions, checks whether guidance was understood, and aggregates misunderstanding patterns for responders.
Project 3	Geospatial Intelligence For Public Health Decisions		5	The main aim of the project is to provide reliable Geospatial Intelligence for Public Health decisions.
Project 4	AI-Powered Disease Risk Prediction System For Zoonotic Disease Outbreaks		5	The aim of this project is to build an AI-powered disease risk prediction system that integrates human, animal, and environmental data to identify and visualize areas at high risk of zoonotic disease outbreaks.

Grading

Component	Weight
Milestone 1: Technical Design and Reproducible Repository	10%
Milestone 2: Proof Of Concept	20%
Midterm oral presentation	10%
Final presentation with a live demo	10%
Project report and code	30%
In-class tests and quizzes (top 5 of the 6 quizzes)	20%

Questions, so far?

Quick poll

Rather have deadlines on Fridays or Sundays?

Our treat

**Our treat: a field trip to the ICRC
Museum in Geneva, *covered by*
*the lab***

Date TBD soon

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Building intelligent Global Health and Humanitarian
Response Technologies

Week 1: *Inequitable Inaccuracy*

Context *Tutti Fratelli*: The Principles That Created a Movement

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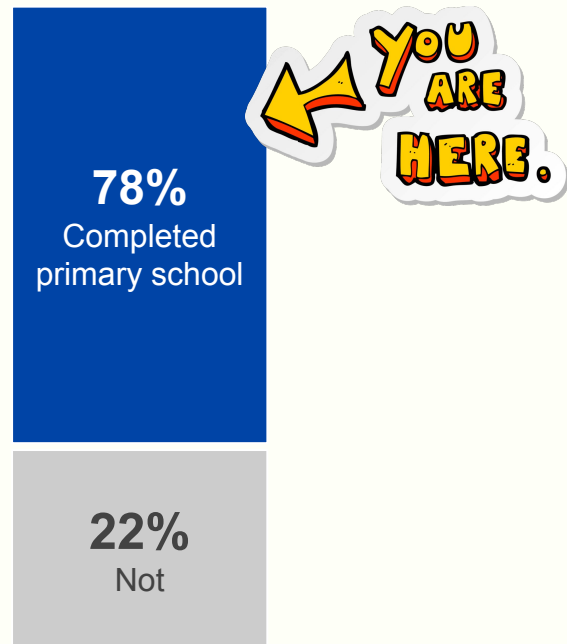
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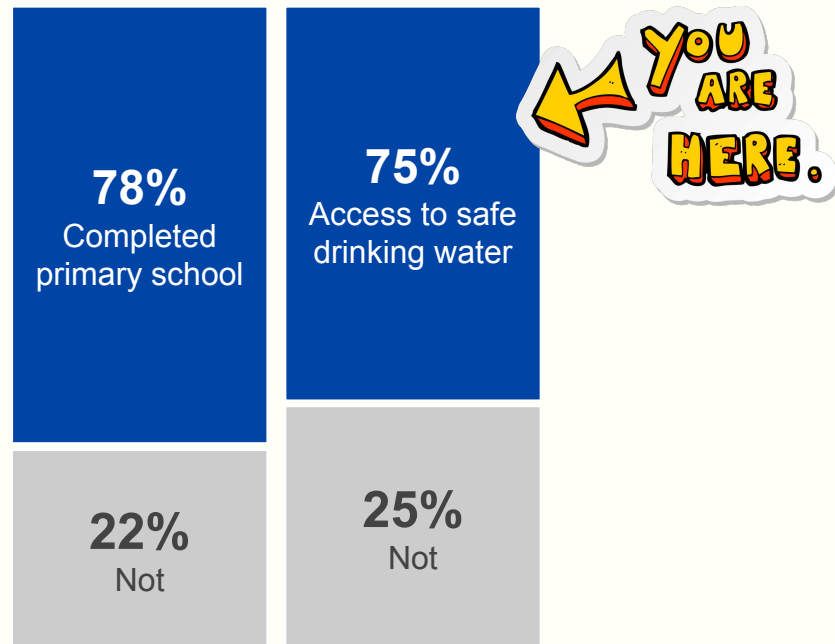
Why do systems *fail* the people who need them most?

- A model that assumes connectivity has already chosen its users.
- A dataset that requires a clinic visit has already excluded the sickest.
- An objective function that maximizes throughput has already decided whose time matters.

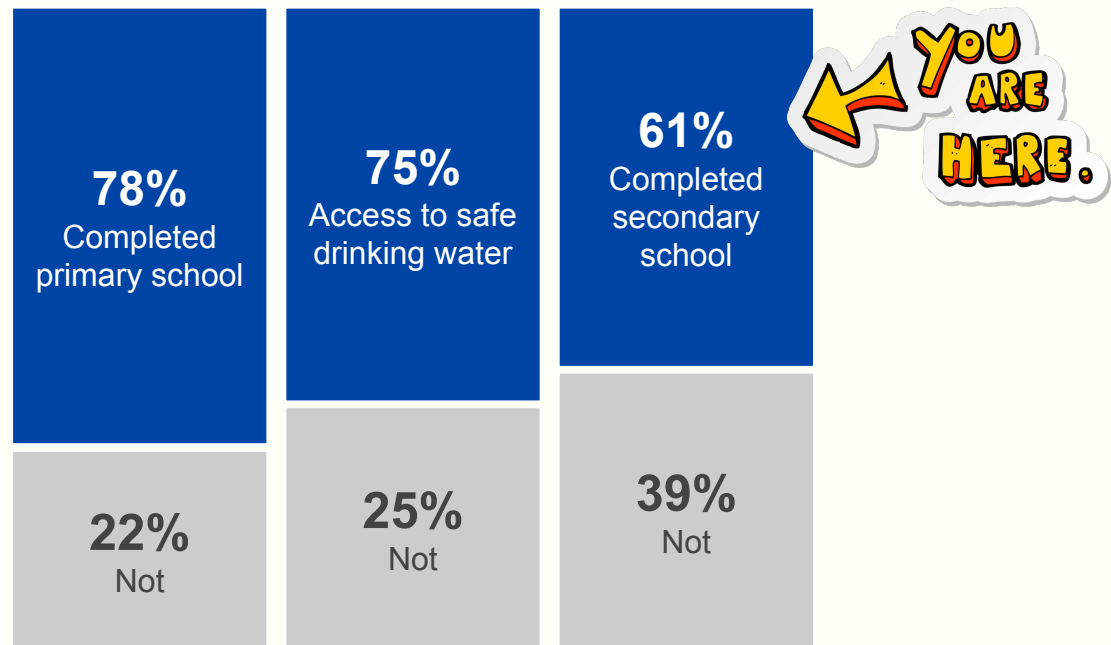
Welcome to the ELITE!



Welcome to the ELITE!



Welcome to the ELITE!



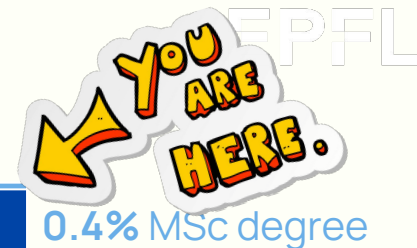
Welcome to the ELITE!



Welcome to the ELITE!

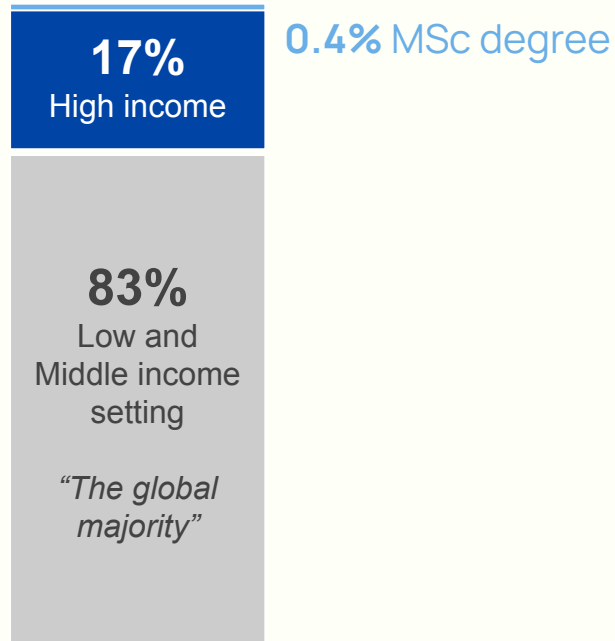


Welcome to the ELITE!

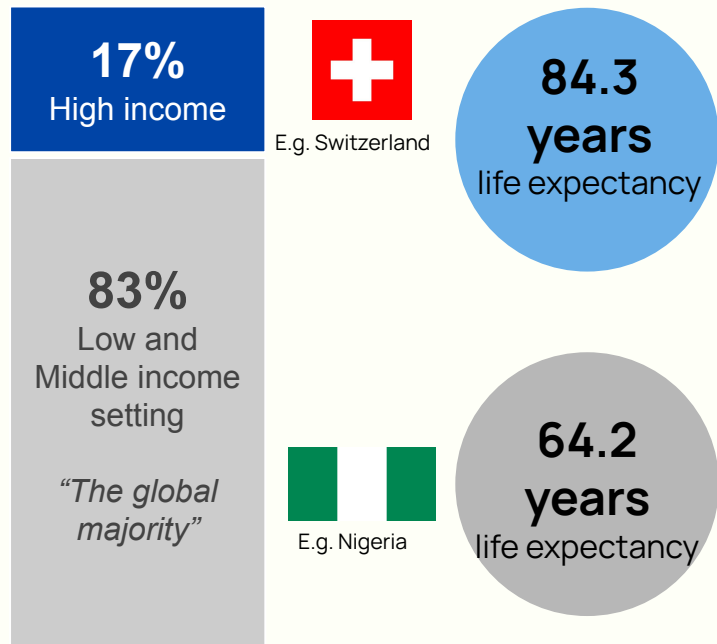


0.4% MSc degree

Welcome to the ELITE! What will you do with this privilege?

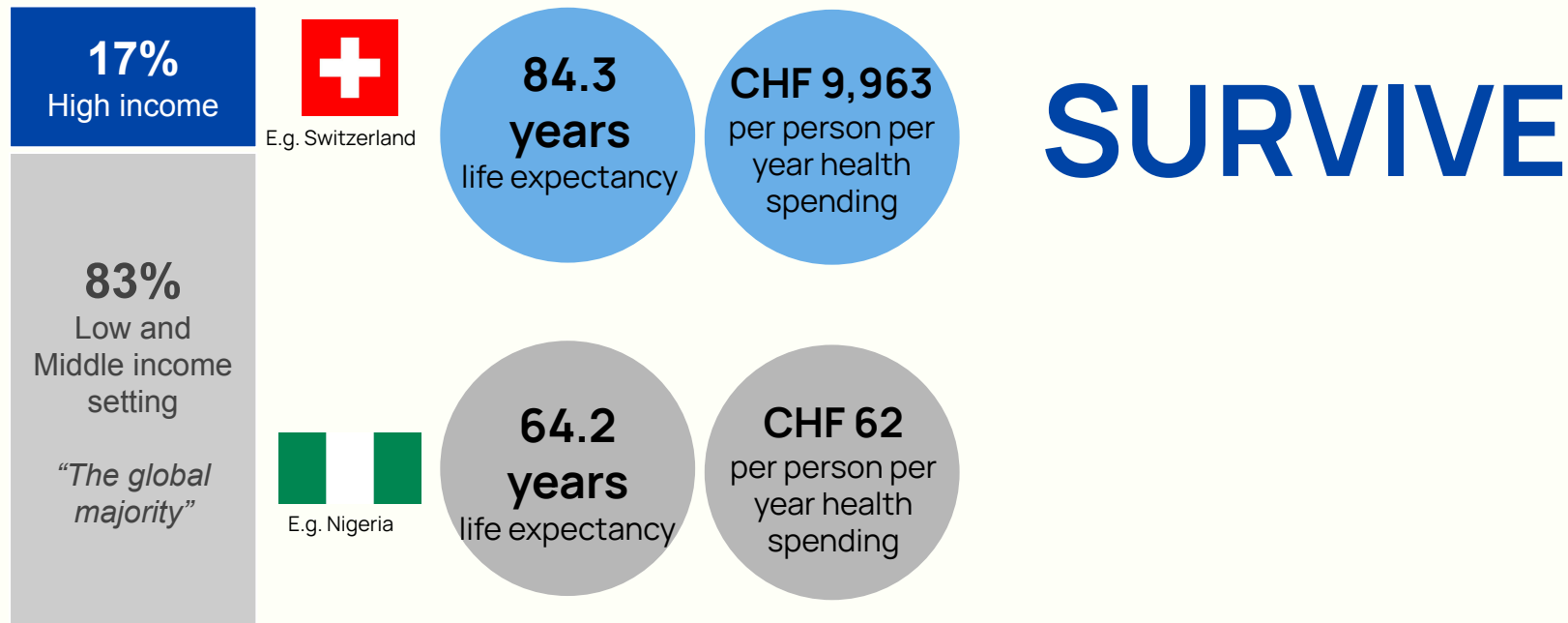


Welcome to the ELITE! What will you do with this privilege?

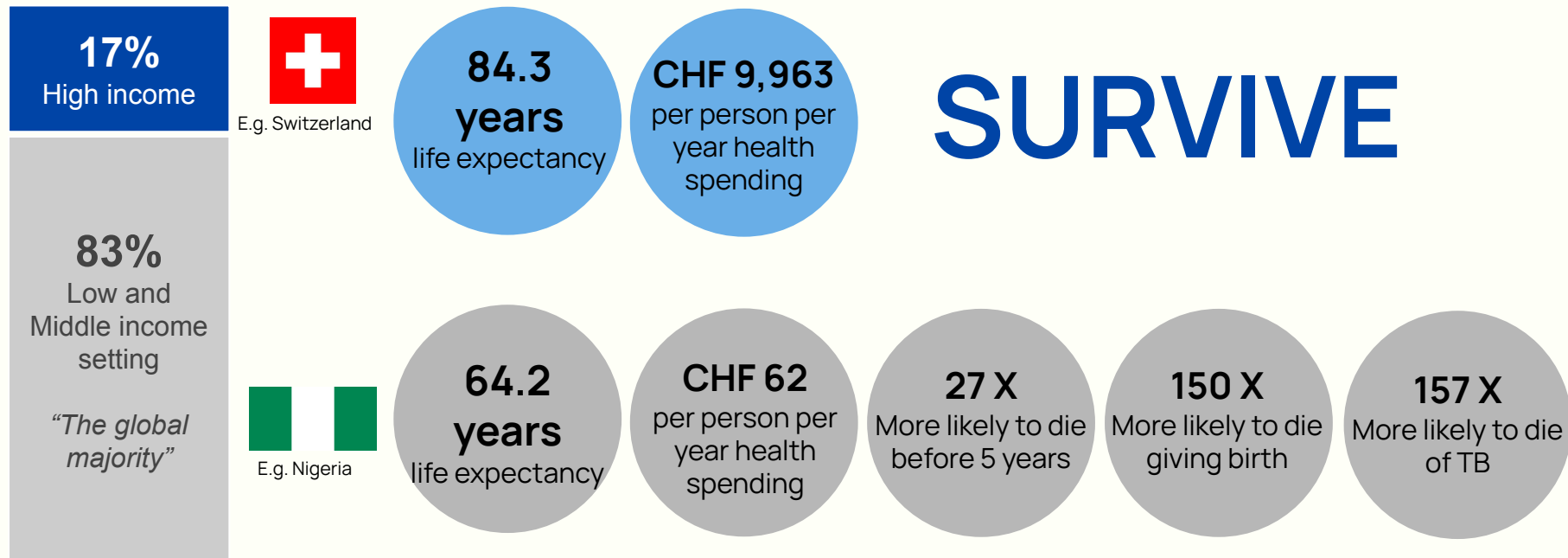


SURVIVE

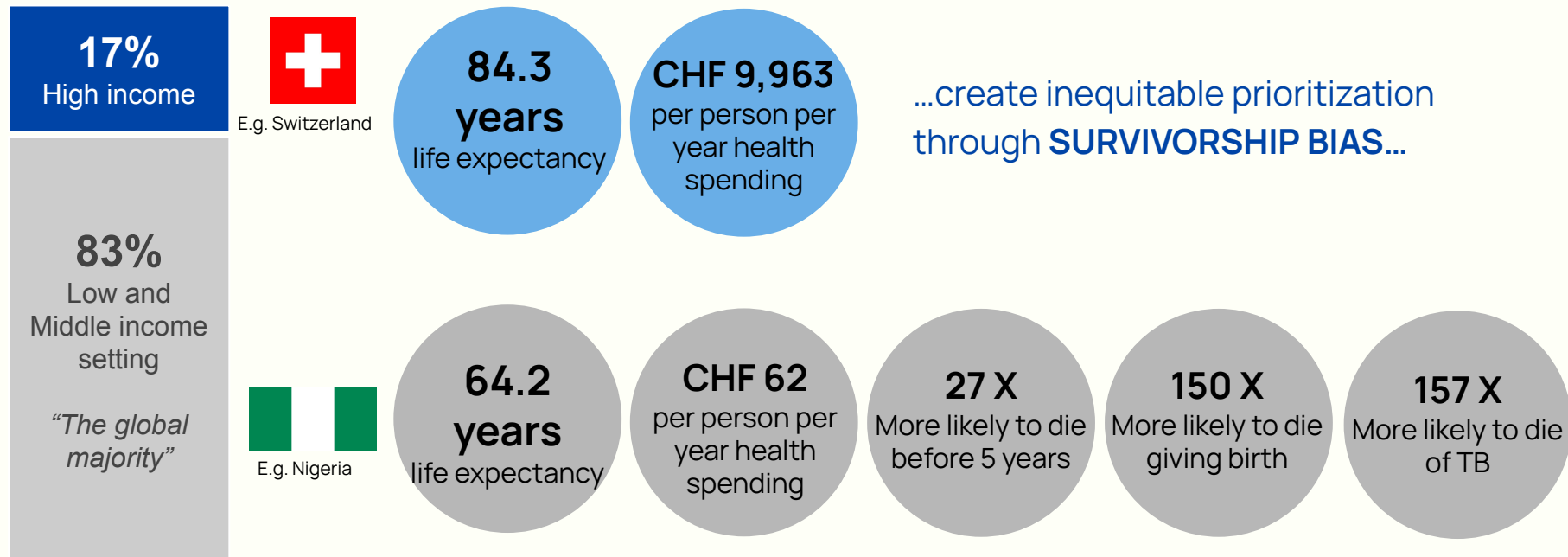
Welcome to the ELITE! What will you do with this privilege?



Welcome to the ELITE! What will you do with this privilege?



Welcome to the ELITE! What will you do with this privilege?



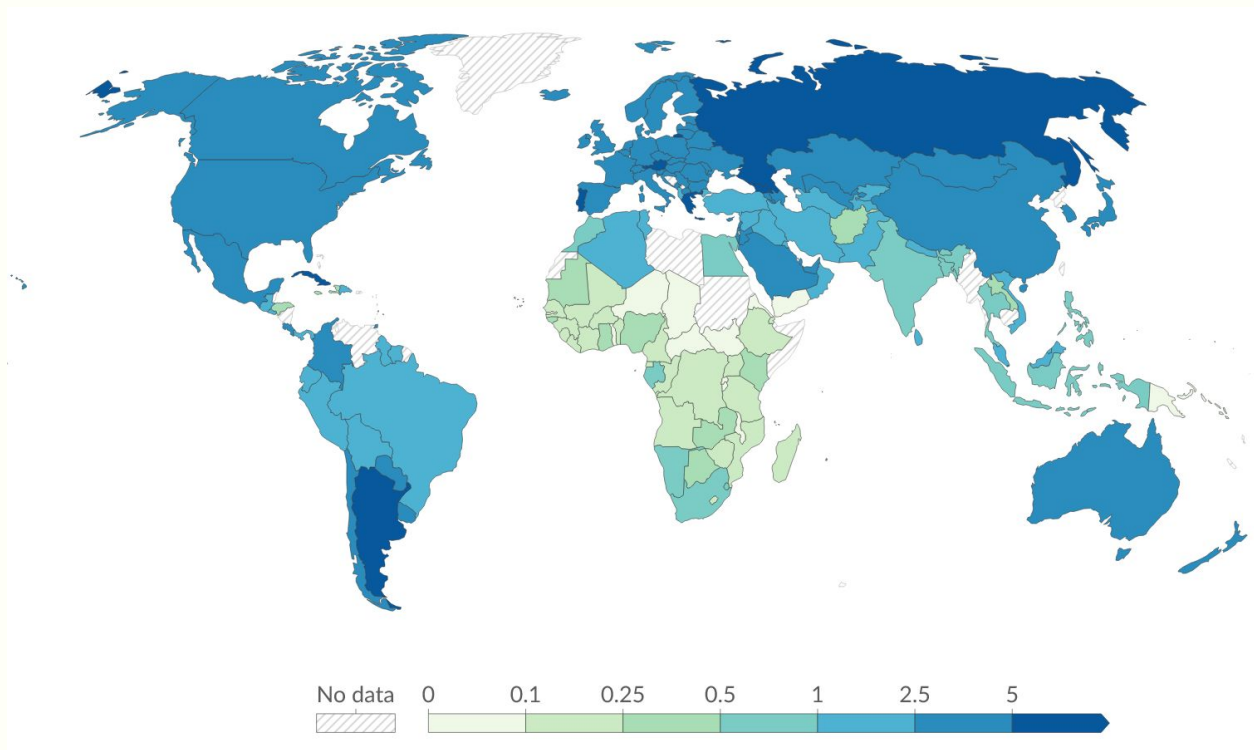
The lottery



1 doctor / 220



1 doctor / 4000



Medical doctors per 1,000 people

The lottery

Gini Index:

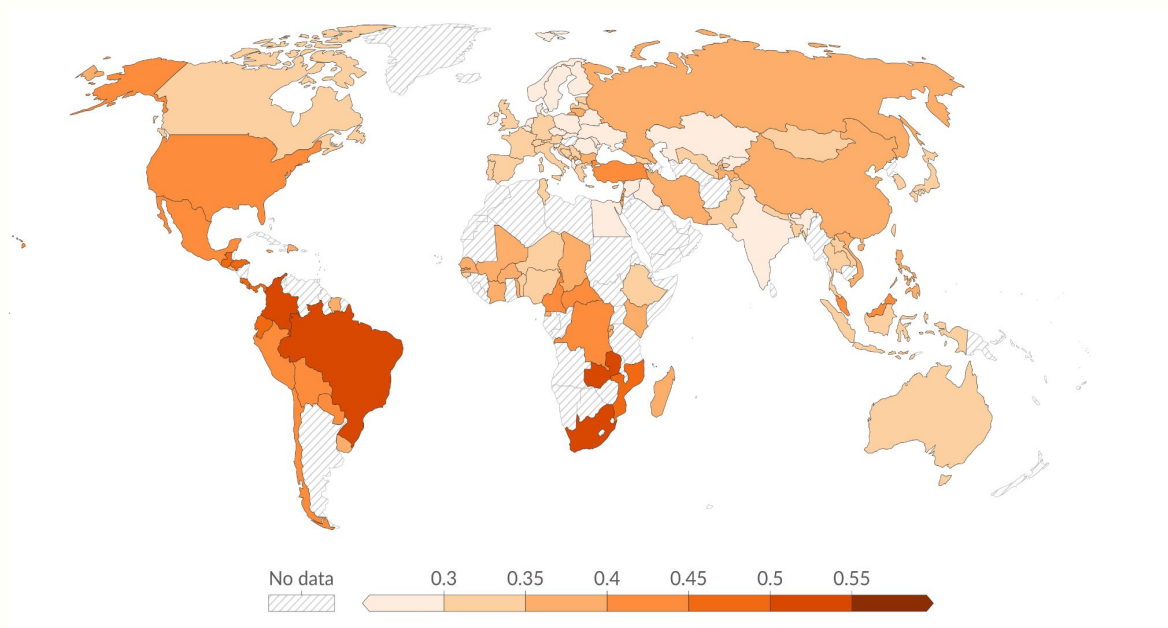
0 (perfect equality)

1 (maximum inequality)



0.6 = Most inequitable country

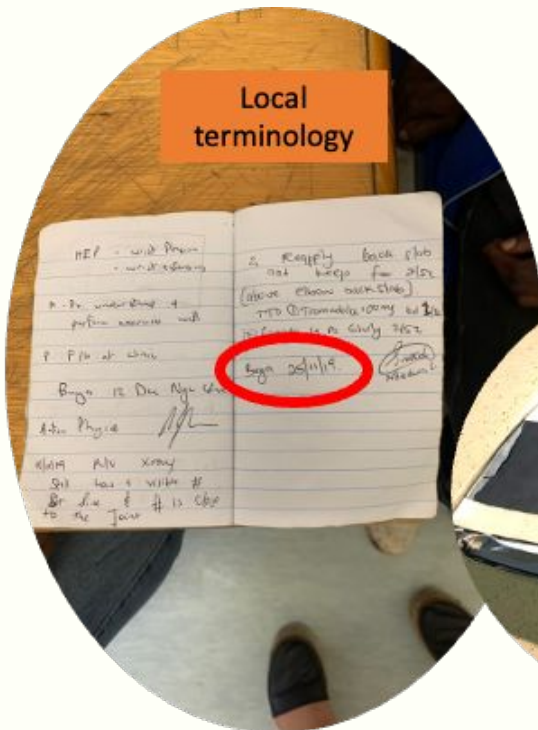
- Richest 10% get 60%
- Poorest 50% get 5%



Data



Patient-owned
health data



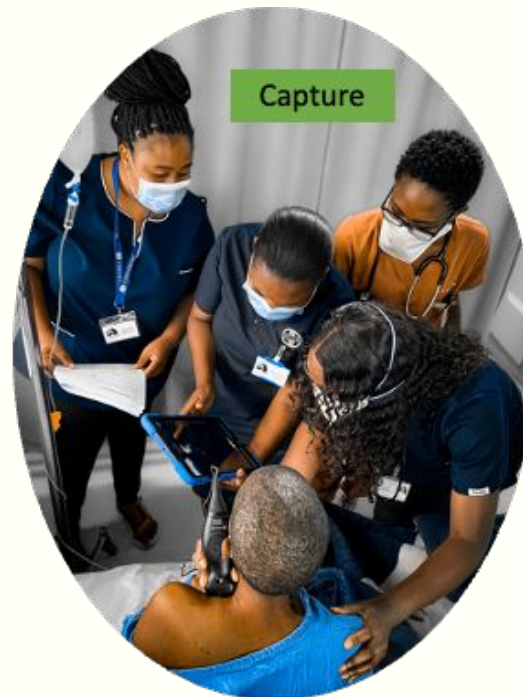
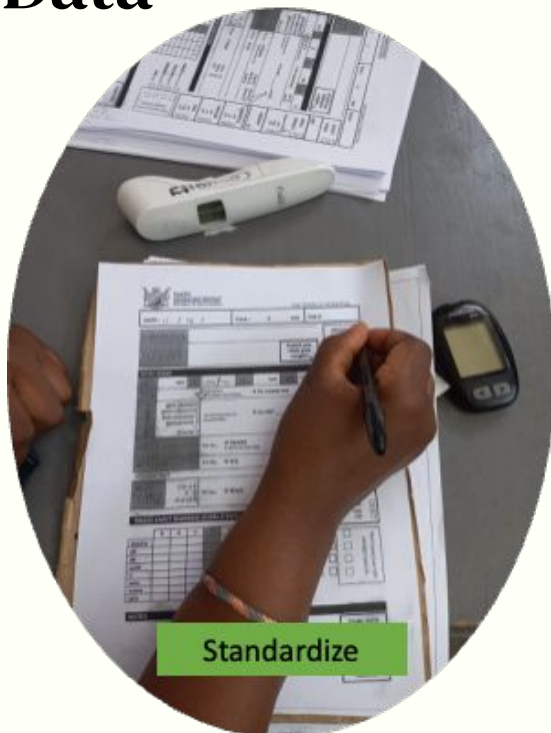
Local
terminology



Reporting to the
government registry

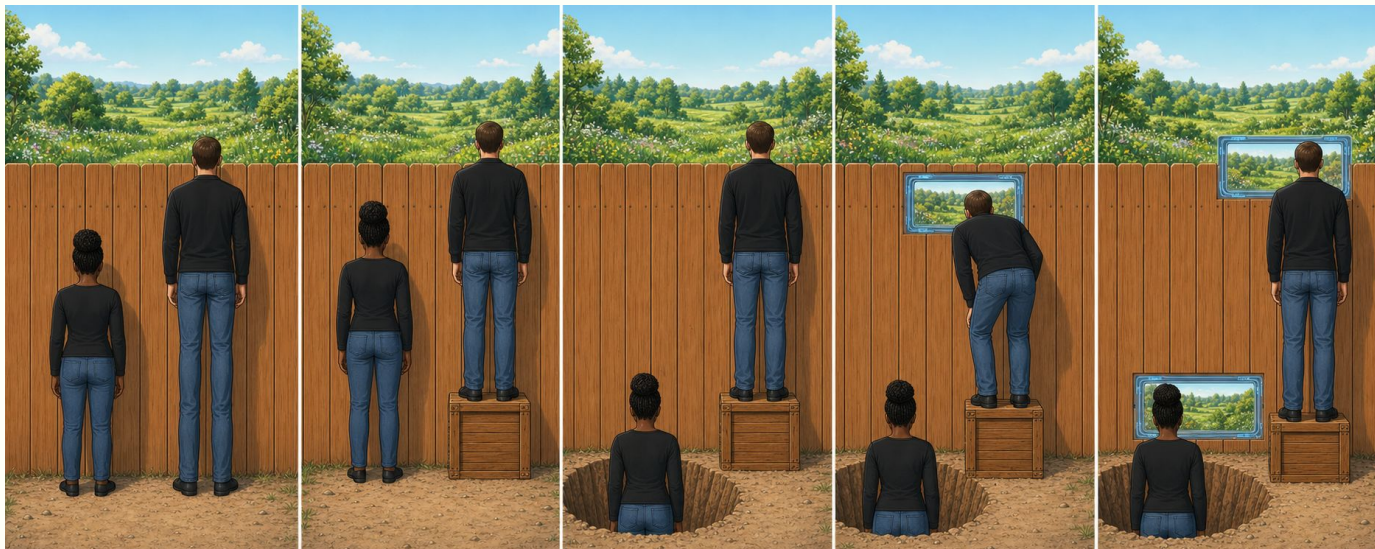
Week 1: An introduction to inequity

Data



Week 1: An introduction to inequity

Equity vs Equality



Innate deficiency /
superiority?

Resource
advantage?

Resource
disadvantage?

Poorly adapted AI

Adapted AI

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Tutti Fratelli: The Principles That Created a Movement

Lecturer: *Prof. Annie Hartley*

Light

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Outline

- 01 The history of humanitarian response
- 02 The humanitarian principles
- 03 The humanitarian system
- 04 Disasters
- 05 Humanitarian standards

Solferino June 24, 1859

CONTEXT

40'000
wounded
soldiers

1 medic /
1000 soldiers

1
veterinarian
/ 100
horses



Henry Dunant

Before the war: Business man from Geneva in the water-mill industry

1859: Travels via Solferino on business...discovers the aftermath

Next days: Organizes relief in the church of Castiglione

Not a medic. Achieves relief through coordination and advocacy

Next years: Founds ICRC

1867: His business goes bankrupt

1901: First Nobel Peace Prize



Founding of the ICRC

1859 (war)

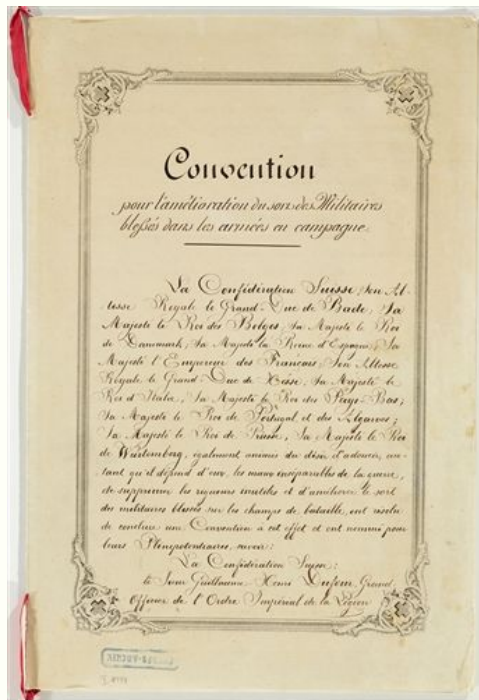
1862 (book)

1863 (ICRC)

1864 (1st Geneva convention)

10 articles (highlights)

- Wounded and sick soldiers must be collected and treated, **regardless of which side they fought for**.
- Medical personnel, ambulances, civilians helping, and military hospitals were declared **neutral** and **protected**.
- The **red cross on a white background** = protective symbol for medical services.



Terminology

Temoignage

“bearing witness”

The humanitarian practice of **speaking out** (publicly or privately) about suffering, violence, neglect, or abuse witnessed during aid work.

Four key terms

Humanity

The claim that the **duty to relieve suffering wherever it is found** overrides other considerations.

Impartiality

No discrimination as to nationality, race, religious belief, class, or political opinion.
Relief is guided solely by need, with priority to the most urgent cases of distress

Neutrality

The Movement **may not take sides** in hostilities or engage at any time in controversies of a political, racial, religious or ideological nature.

Independence

Humanitarian action must remain **autonomous** from political, economic, or military objectives. Able to act on the principles even against the interests of donors and host states.

Applied to AI?

Humanity

Purpose-driven

Use AI only when it protects life, relieves suffering, and preserves human dignity. Efficiency must never outweigh foreseeable harm to people.

Impartiality

Fairness

AI-supported decisions must be based on need alone. Not identity, status, data availability or politics. Test data and outputs for bias, exclusion, and unequal errors.

Neutrality

Interpretability

Humanitarian AI must not take sides or support military targeting, partisan surveillance, propaganda, or political objectives that undermine trust.

Independence

Control

Governments, donors, militaries, and technology companies must not determine whom AI serves. Humanitarian actors must retain meaningful control and the power to question, override, refuse, or stop its use.

Dilemmas

- **Neutrality** may require restraint to preserve trust and access
- **Humanity** may require speaking out against severe suffering
- **Impartiality** may require giving more assistance to those in greatest need even when this looks like taking sides.

Terminology

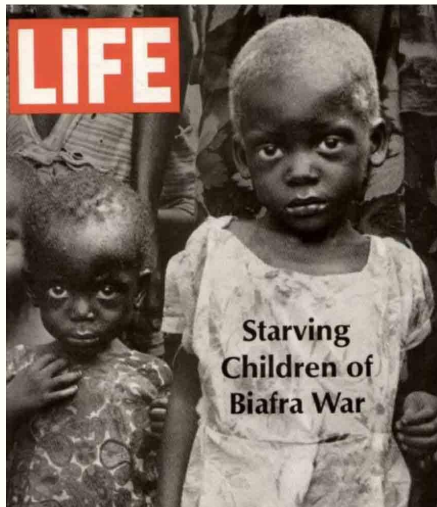
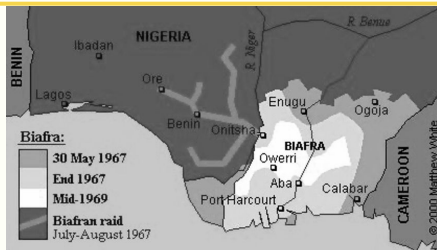
Dilemma

A situation in which two legitimate duties conflict, so that protecting one requires compromising another. No available choice fully satisfies every principle.

Dilemmas that shaped the humanitarian system

1969 (Biafra war)

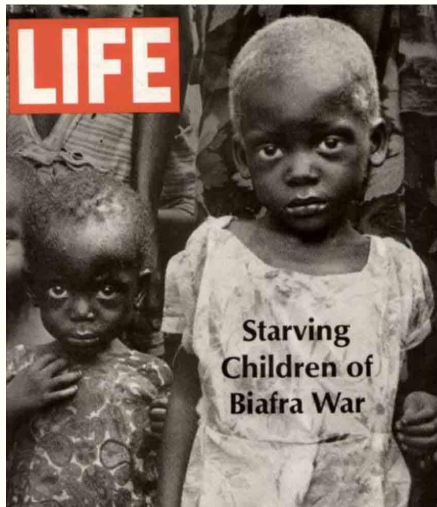
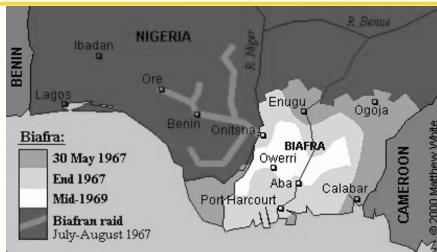
Biafra succeeds from Nigeria
→ Encircled by Nigerian forces
→ **Massive famine (2M died)**



Dilemmas that shaped the humanitarian system

1969 (Biafra war)

Biafra succeeds from Nigeria
→ Encircled by Nigerian forces
→ **Massive famine (2M died)**



Dilemma: Access vs Protest

- 1) **Access as protection**
Remain present, negotiate confidentially and preserve access to victims.
- 2) **Protest as protection**
Treat victims, bear public witness and mobilise pressure against intolerable suffering.



ICRC



Dilemmas that shaped the humanitarian system

1969 (Biafra war)



1980 (Vietnam boat people)



Communist victory in Vietnam

→ 100k+ fled to sea ... but no country let them in as refugees

→ A ship with 2K people onboard were starving

Dilemmas that shaped the humanitarian system

1969 (Biafra war)



1980 (Vietnam boat people)



Communist victory in Vietnam

→ 100k+ fled to sea ... but no country let them in as refugees

→ A ship with 2K people onboard were starving

Dilemma: Dialogue vs Action

- 1) **Diplomacy as protection**
Negotiate with states to secure safe landing (but leave leave at risk).
- 2) **Direct action as protection**
Go to those stranded at sea, bear witness and pressure governments to act.



Dilemmas that shaped the humanitarian system

1969 (Biafra war)



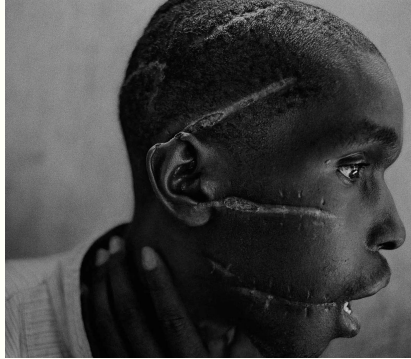
1980 (Vietnam boat people)



1994 (Rwandan Genocide)



People killed
6/minute
330/hour



1M killed in 100 days, 2M fled into neighbouring countries

→ Overcrowded camps lacked sufficient water, sanitation and healthcare

→ Armed groups diverted humanitarian assistance

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Dilemmas that shaped the humanitarian system

1969 (Biafra war)



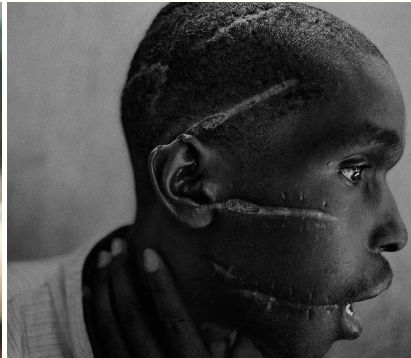
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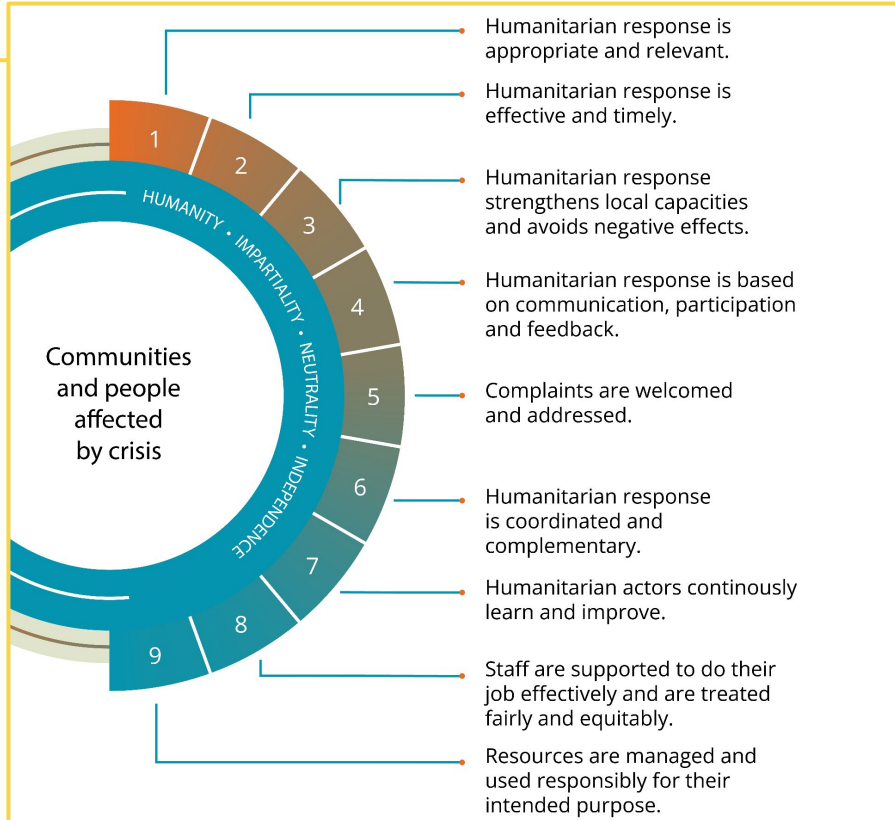
1M killed in 100 days, 2M fled into neighbouring countries
→ Overcrowded camps lacked sufficient water, sanitation and healthcare
→ Armed groups diverted humanitarian assistance

Dilemma: Is any aid better than no aid?

- 1. Urgency as protection**
Deliver whatever assistance is available now –regardless of quality
- 2. Standards as protection**
People have a right to competent and coordinated assistance. Agencies must meet minimum standards and remain accountable for the effects of their aid.



Sphere standards



1994 (Rwanda Genocide)



Humanitarian Charter

+

Measurable Minimum Standards

To ensure sustainable survival and dignity

Sphere standards...what is “minimum”?

WATER



SHORT-TERM EXCEPTION

7.5 L

per person / day
Acute drought • short time only

MAXIMUM DISTANCE

≤ 500 m

From every household
to a water point



SANITATION



RAPID-ONSET

1 : 50

Communal • improve
as soon as possible

MAXIMUM DISTANCE

≤ 50 m

Dwelling to shared toilet

MEDIUM-TERM RATIO

3 : 1

Female : male toilets

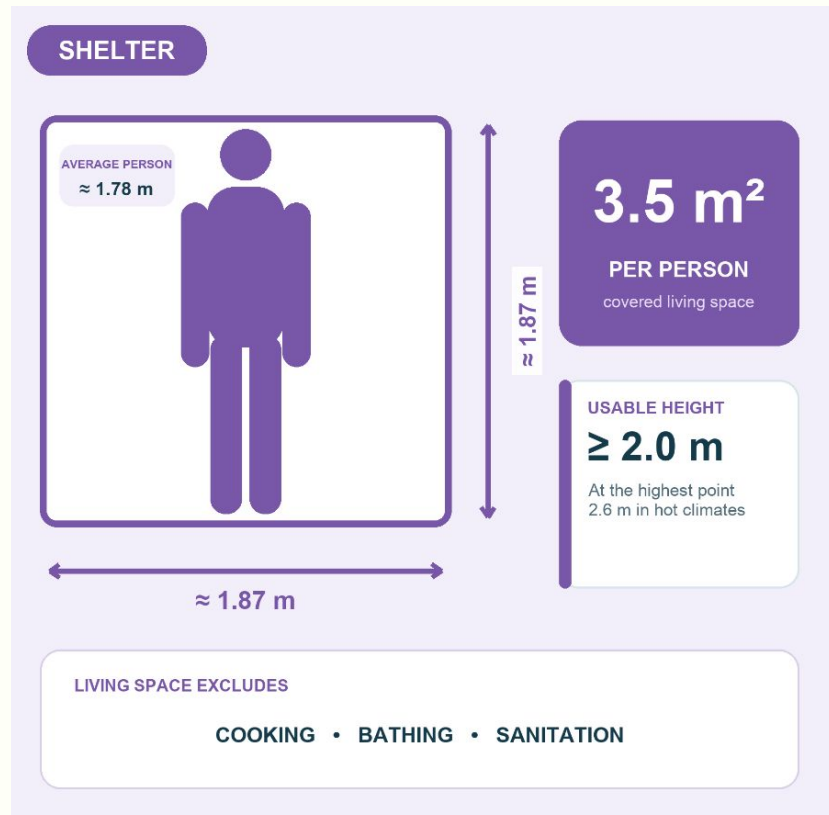
1 TOILET : 20 PEOPLE



SAFETY

Inside locks + adequate lighting
for every shared toilet

Sphere standards...what is “minimum”?



Disaster

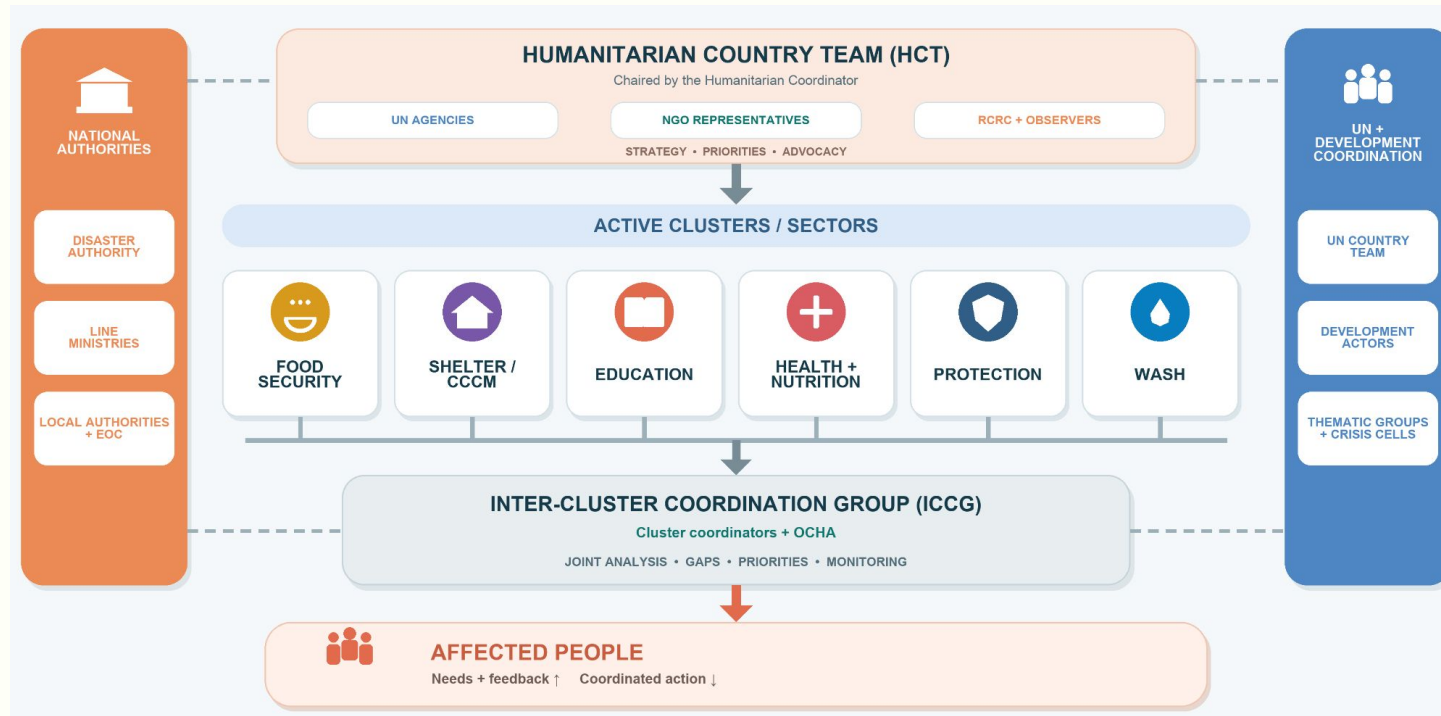
$$\text{Impact} = \frac{\text{Hazard} \times \text{Exposure} \times \text{Vulnerability}}{\text{Capacity}}$$

Terminology

Disaster

A serious disruption of a community's functioning that exceeds its capacity to cope using its own resources.

Coordination



Terminology

Clusters

The UN-led coordination model that divides response into **sectoral clusters**

- Health (WHO)
- Shelter
- Nutrition
- WASH
- Protection
- ...

Data protection is protection

150M records since 1914

Persistence = value

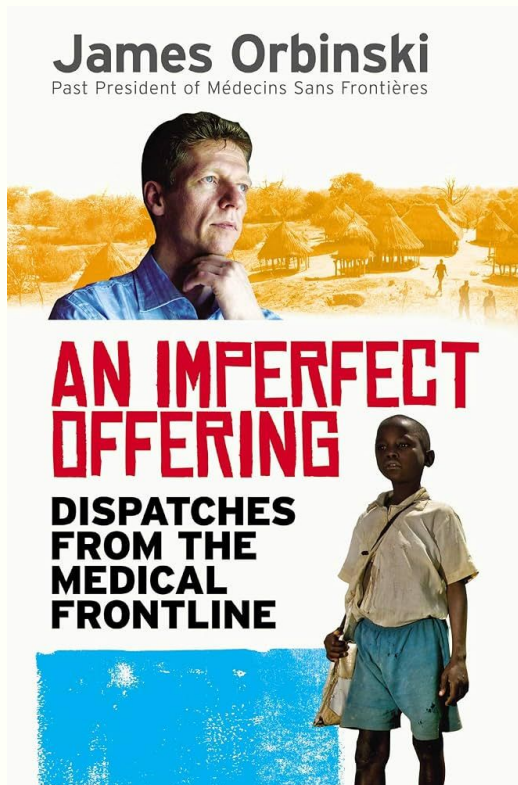
...but also lengthens lifetime of exposure risk



Connecting to AI

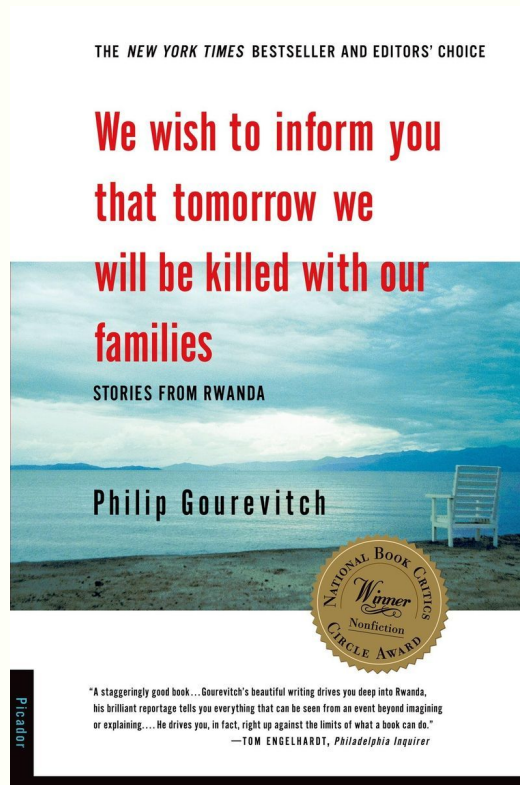
DATA MINIMIZATION

Collect **only what the stated purpose requires**, keep it only as long as needed, at the coarsest resolution that still serves the purpose.
(Core principle of the ICRC *Handbook on Data Protection in Humanitarian Action*.)



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End

DOMAIN

Questions?

Light

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Thank You

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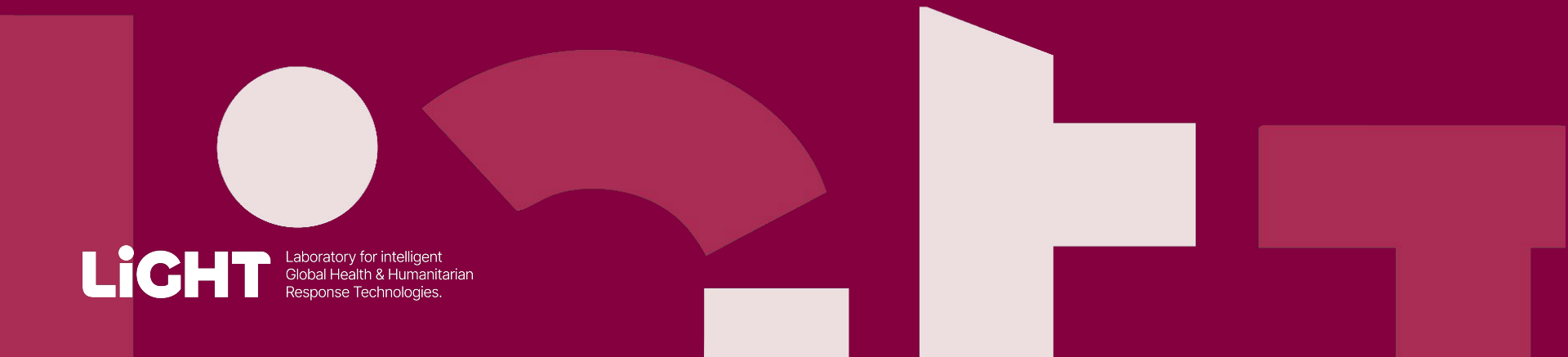


Building for *Broken* Environments

Lecturer: *Dr. Lars Klein*

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Engineering

This course will be heavily project-driven

Imagine that you're building for people in need, that your engineering affects them immediately

One key term here is **Accountability**

How do we prevent mistakes?

Why does this matter so much?

In the Wild...

Let's assume you're a bio-informatician...

Human genes

MARCH1
SEPT2
DEC3

Mouse RIKEN IDs

2310009E13
0610009E02

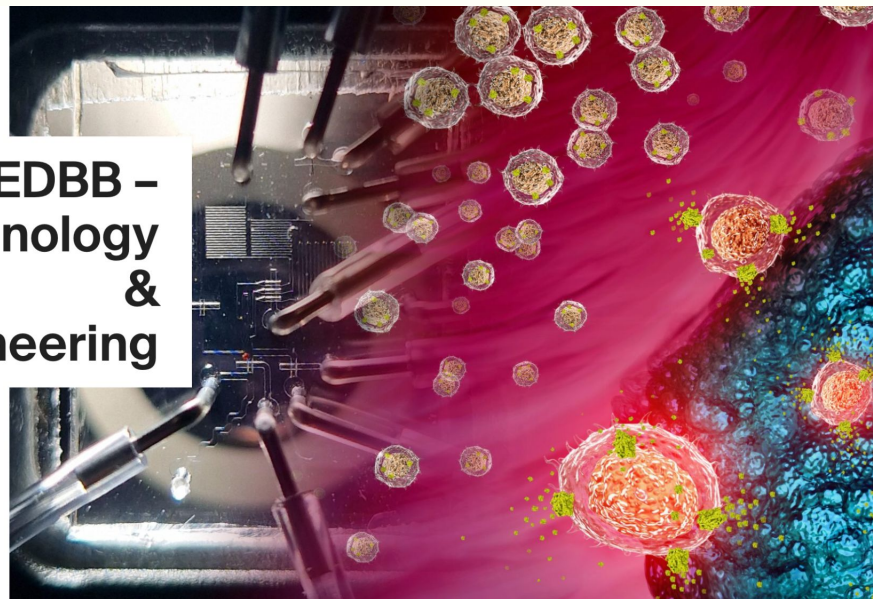
AND
OR
NOT

Drosophila genes

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**EDBB –
Biotechnology
&
Bioengineering**



In the Wild...

Mistaken Identifiers: Gene name errors can be introduced inadvertently when using Excel in bioinformatics [paper](#)

Background *When processing microarray data sets, we recently noticed that some gene names were being changed inadvertently to non-gene names.*

Results *A little detective work traced the problem to default date format conversions and floating-point format conversions in the very useful Excel program package. The date conversions affect at least 30 gene names; the floating-point conversions affect at least 2,000 if Riken identifiers are included. These conversions are irreversible; the original gene names cannot be recovered.*

In the Wild...

Gene name errors are widespread in the scientific literature [paper](#)

*Of the selected journals, the proportion of published articles with Excel files containing gene lists that are **affected by gene name errors is 19.6 %**. [...]*

Journals that had the highest proportion of papers with affected supplementary files were Nucleic Acids Research, Genome Biology, Nature Genetics, Genome Research, Genes and Development and Nature (>20 %)

In the Wild...

Guidelines for human gene nomenclature [paper](#)

Box 3 Scenarios that may merit a symbol change

...

- **Symbols that affect data handling and retrieval.** For example, all symbols that autoconverted to dates in Microsoft Excel have been changed (for example, *SEPT1* is now *SEPTIN1*; *MARCH1* is now *MARCHF1*); tRNA synthetase symbols that were also common words have been changed (for example, *WARS* is now *WARS1*; *CARS* is now *CARS1*).

[Show less](#) ^

In the Wild...

Gene name errors: Lessons not learned [paper](#)

Although recent changes have been made to human and mouse gene names to prevent conversion to dates [3], it is uncertain whether such changes have propagated through to other species. To assess this, we downloaded all eukaryotic gene names available in Ensembl and imported this into Excel and collected all genes that were converted to dates

The proportion of publications with Excel gene lists that contain errors was 30.9%; substantially higher than previously reported

So now what?

[Commentary blog post from the authors](#)

Our research shows autocorrect errors, particularly in Excel spreadsheets, can also make a mess of gene names in genetic research. We surveyed more than 10,000 papers with Excel gene lists published between 2014 and 2020 and found [more than 30%](#) contained at least one gene name mangled by autocorrect.

This research follows our 2016 study that found [around 20%](#) of papers contained these errors, so the problem may be getting worse.

We believe the lesson for researchers is clear: it's past time to stop using Excel and learn to use more powerful software.

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LiGHT

EPFL

ARIADNE LABS

ASHOKA
UNIVERSITY

WORLD
ECONOMIC
FORUM

Rwanda
Centre for the
Fourth Industrial
Revolution



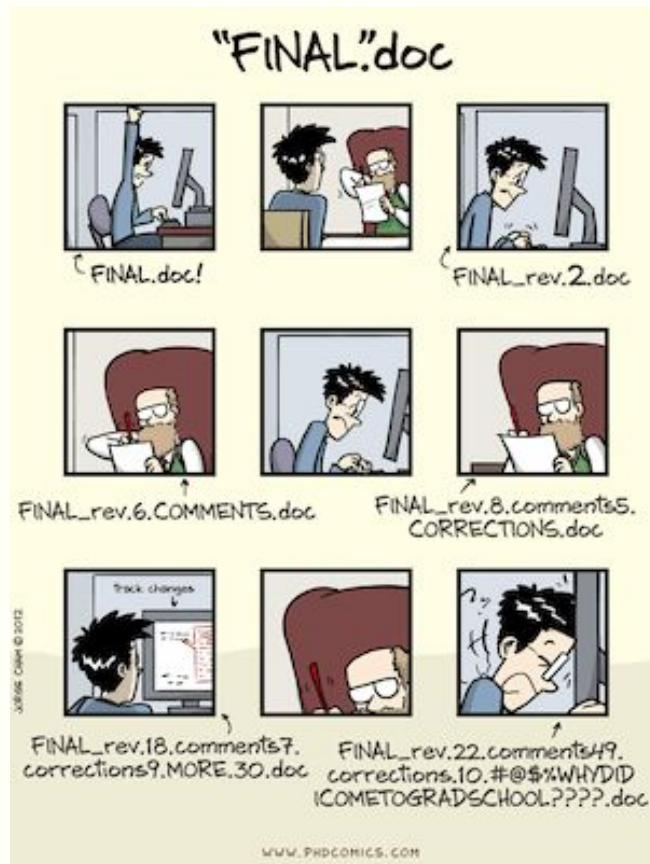
Chances are you already do



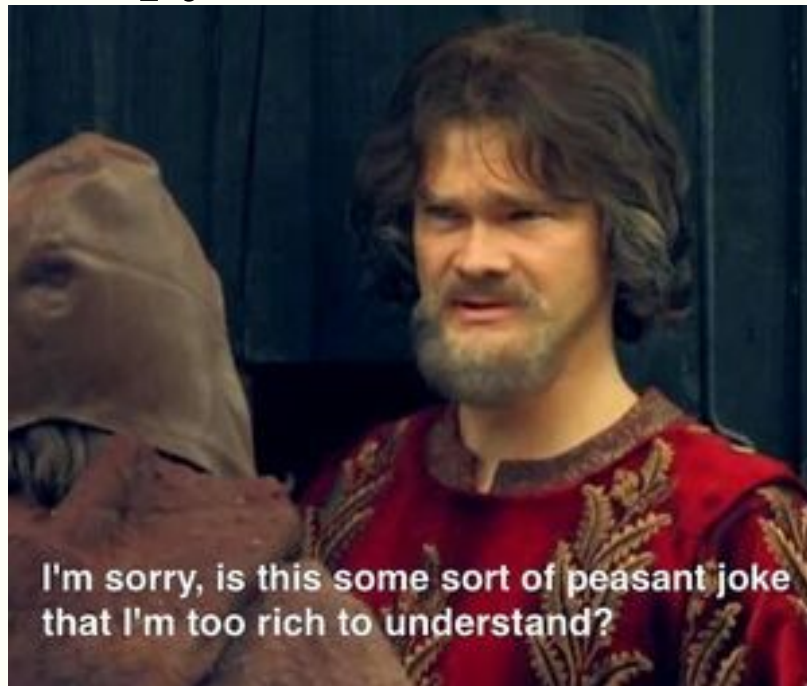
Outline

- 01 Git
- 02 Code Hygiene
- 03 Reproducibility
- 04 Tracking
- 05 Project Time!

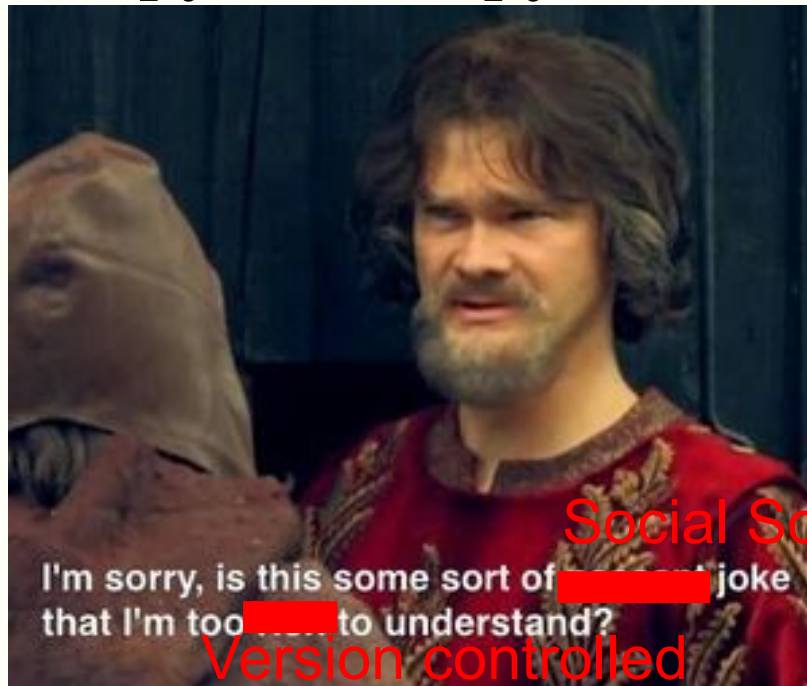
final_v2_ACTUAL.py



final_v2_ACTUAL.py



~~final_v2_ACTUAL.py~~ -> final.py



Social Sciences

LiGHT

Laboratory for intelligent
Global Health & Humanitarian
Response Technologies.

final.py

applying feedback from ...

1 minute ago



Social Sciences

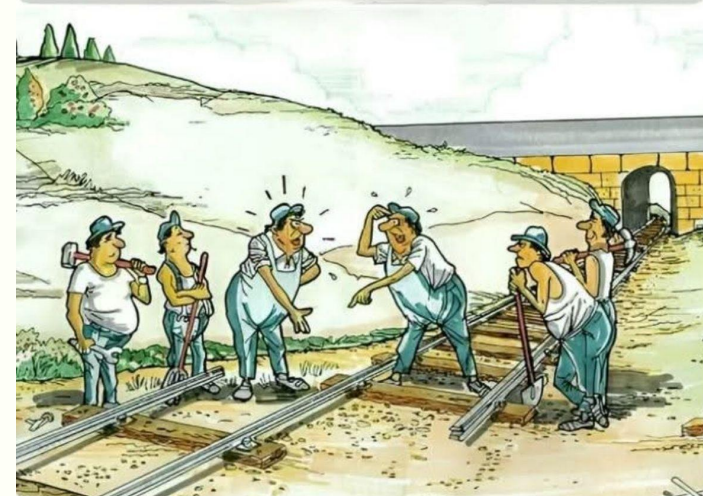
Git

Please ask yourself:
What is the LARGEST team size you've ever worked in?

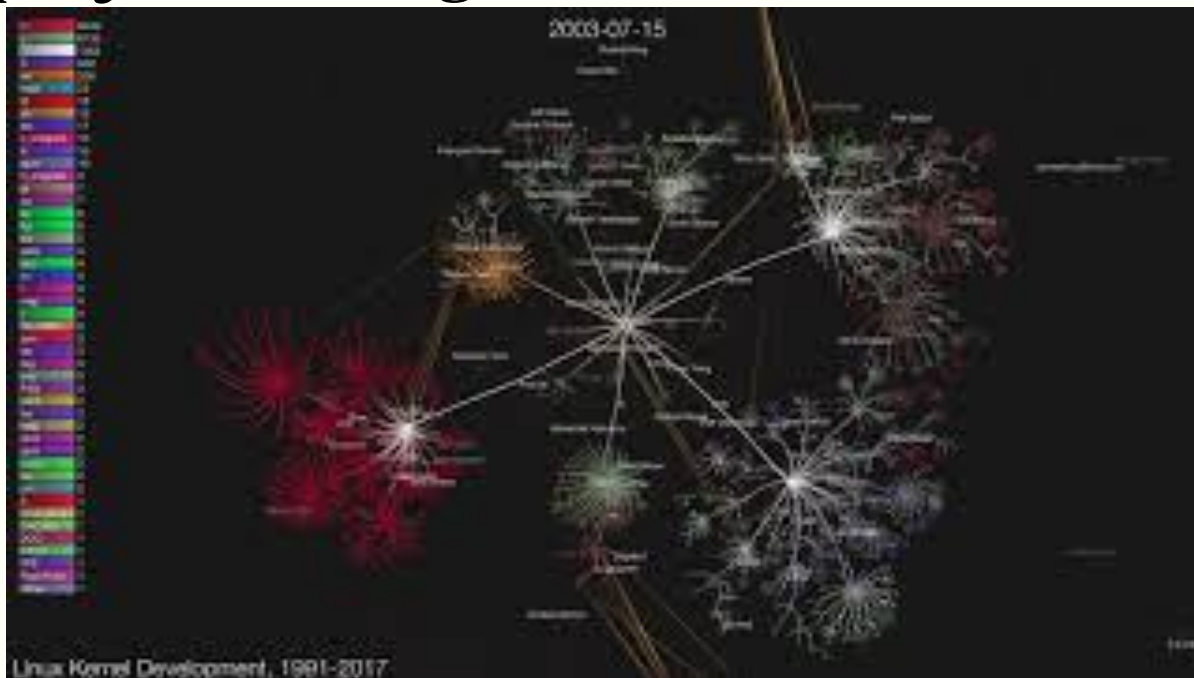
Did that work well?

How much did you need to coordinate with your teammates?

**"TEAMWORK WITHOUT
COORDINATION LEADS
TO CONFUSION"**



LARGE projects with git ...



Git sanity checklist

All your projects MUST live on git. **ToDo**, point to light org here.
This checklist is grade relevant.

Git sanity checklist

All your projects MUST live on git. ToDo, point to light org here.
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1. Never edit the main branch directly. Development happens on feature branches

Managing a branch protection rule

You can create a branch protection rule to enforce certain workflows for one or more branches, such as requiring an approving review or passing status checks for all pull requests merged into the protected branch.

Who can use this feature?

- 🔑 People with admin permissions or a custom role with the "edit repository rules" permission to a repository can manage branch protection rules.
- 📦 Protected branches are available in public repositories with GitHub Free and GitHub Free for organizations. Protected branches are also available in public and private repositories with GitHub Pro, GitHub Team, GitHub Enterprise Cloud, and GitHub Enterprise Server. See [GitHub's plans](#).

Git sanity checklist

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1. Protect the main branch. Development happens on feature branches
2. Branches must be given a semantic name



dev_lars_rangein...



21 Branches

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draft: Dev/lars/fa2layout #16 

 Merged

lhk merged 14 commits into `main` from `dev/lars/fa2layout`  on Nov 24, 2025

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This is a shared responsibility. You must empower your teammates to collaborate with you.

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6. Data, model weights/checkpoints, .env files, keys don't go in git, write .gitignore on day 1

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```
[(base) 1klein@Larss-MacBook-Pro push2whisper % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Code Hygiene

We do expect you to review each other's PRs.

But no one wants to review something like this:

Add black and pre-commit hooks, apply formatting

 [angband-env](#)

1 parent [3d724a4](#) commit 4e453e9 

34 files changed +1,630 -859 

Code Hygiene

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What was the problem in the previous message?

34 changes files, ~1600 lines of code added, ~800 removed!

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Your commits **MUST**
be semantic.

Each change is **self
contained**, each
change is **justified**,
each change is
documented.

Code Hygiene

YAML

 Copy

```
repos:
  - repo: https://github.com/astral-sh/ruff-pre-commit
    rev: v0.16.1
    hooks:
      - id: ruff-check
        args: [--fix]
      - id: ruff-format
```

Reproducibility

Please ask yourself:

Did you ever wonder: How did this work? What numbers went into this plot?

Why can I not get the same results again?

What if patient-facing decisions depend on your numbers?

Reproducibility

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What if patient-facing decisions depend on your numbers?

Survey sheds light on the 'crisis' rocking research.

More than 70% of researchers have tried and failed to reproduce another scientist's experiments, and more than half have failed to reproduce their own experiments. Those are some of the telling figures that emerged from *Nature's* survey of 1,576 researchers who took a brief online questionnaire on reproducibility in research.

Reproducibility

So many reasons reproducibility can fail ...

November 19, 2021

Builds Today, Breaks Tomorrow: The Mystery of the Disappearing NPM Dependency

Bella Wiseman, Lead Engineer, Open Source Program Office

Reproducibility

So many reasons reproducibility can fail ...

```
gatsby
|_gatsby-cli
  |_gatsby-recipes
    |_@graphql-tools/schema
      |_value-or-promise
        |_@changesets/cli
          |_tty-table
            |_smartwrap
```



Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

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Easier said than done. For dependencies there's lockfiles

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But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

`package-lock.json` is automatically generated for any operations where npm modifies either the `node_modules` tree, or `package.json`. It describes the exact tree that was generated, such that subsequent installs are able to generate identical trees, regardless of intermediate dependency updates.

This file is **intended to be committed** into source repositories, and serves various purposes:

Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

Easier said than done. For dependencies there's lockfiles.

But how do you track logic, configs, etc?

Reproducibility

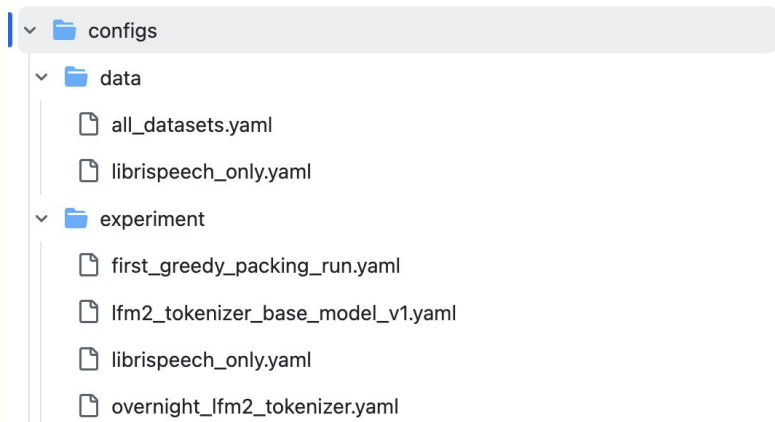
Many pieces to this puzzle.

But one may be particularly important:

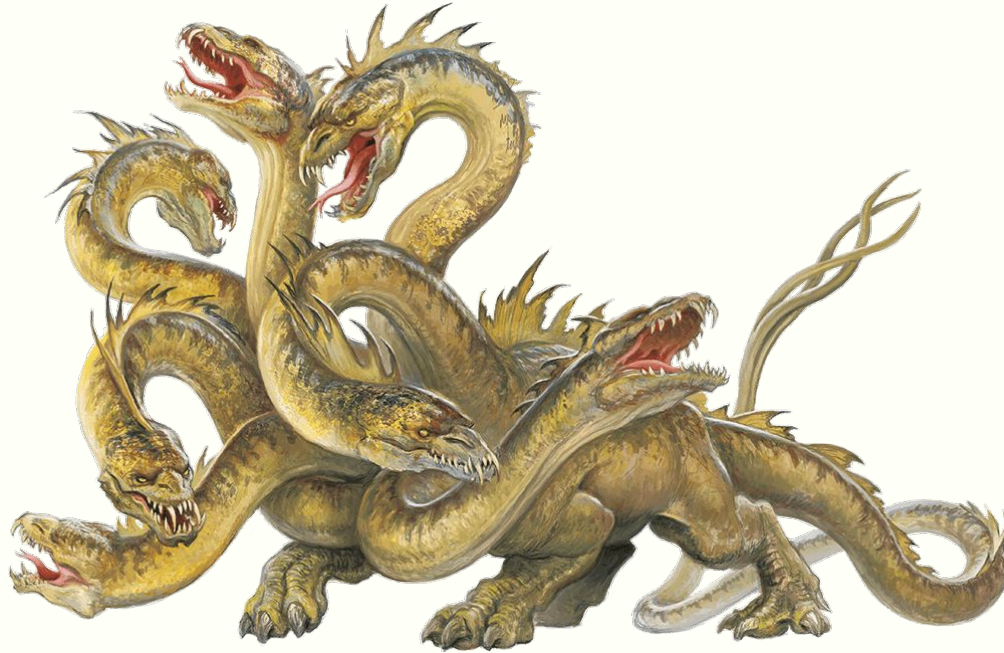
Keep very careful track of all the parameters that go into an experiment!

Easier said than done. For dependencies there's lockfiles.

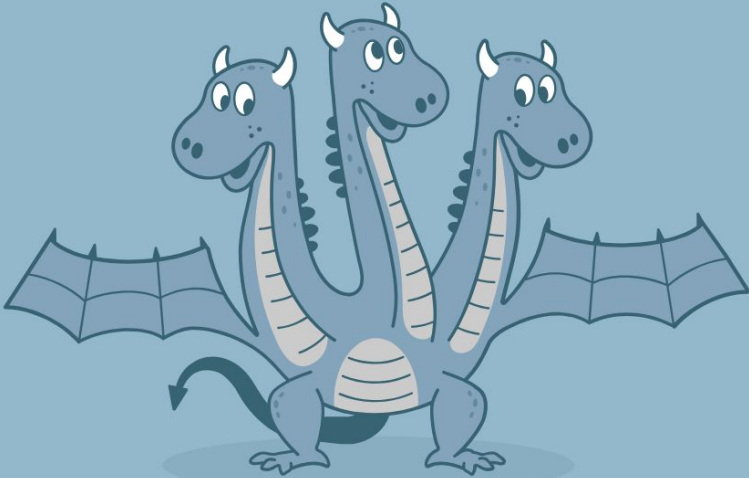
But how do you track logic, configs, etc?



Hydra



Also Hydra (only mildly less scary when first encountered)



Hydra

A framework for elegantly configuring complex applications

[Get Started](#)

 Star 10,602

Tracking & Logging

If a tree falls in a forest and no one is around to hear it, does it make a sound?

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Golden Rule:

You **MUST** account for the resources you use.

→ Every number you produce, **MUST** be reproducible (if it isn't, the compute was wasted!)

But how?

W&B

- **Unified dashboard:** Central repository for all your model metrics and predictions.
- **Lightweight:** No code changes required to integrate with Hugging Face.
- **Accessible:** Free for individuals and academic teams.
- **Secure:** All projects are private by default.
- **Trusted:** Used by machine learning teams at OpenAI, Toyota, Lyft, and more.

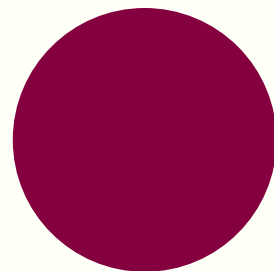
W&B

W&B works like GitHub for machine learning models. Save machine learning experiments to your private, hosted dashboard. Experiment with the confidence that W&B saves all versions of your models, no matter where you run your scripts.

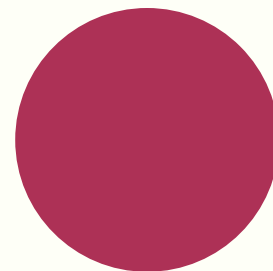
W&B lightweight integrations work with any Python script. Sign up for a free W&B account to start tracking and visualizing your models.

In the Hugging Face Transformers repository, W&B has instrumented the Trainer to automatically log training and evaluation metrics to W&B at each logging step.

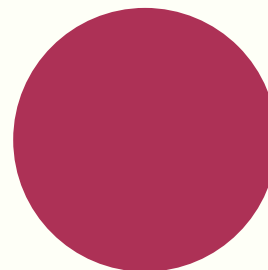




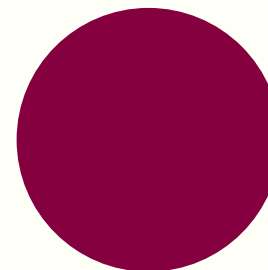
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Title



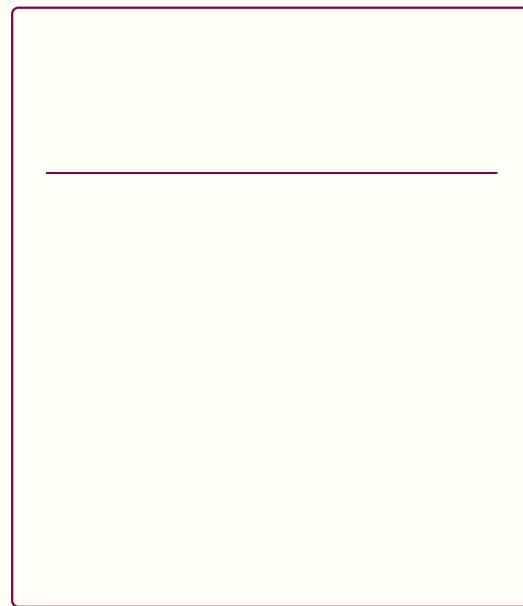
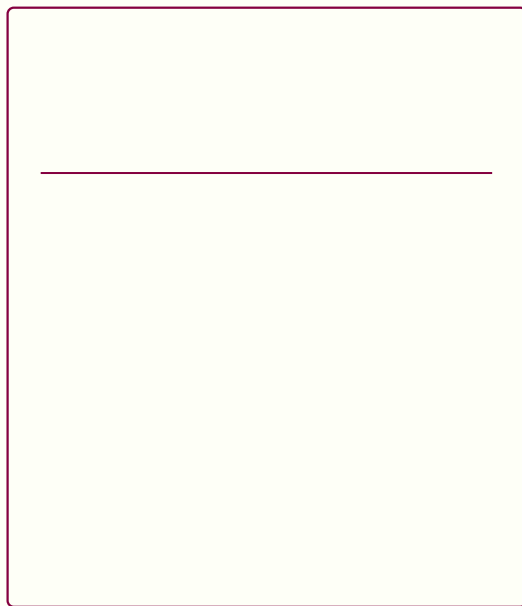
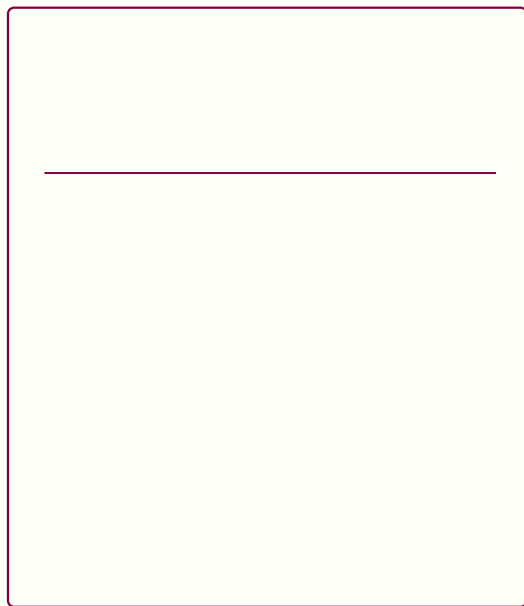
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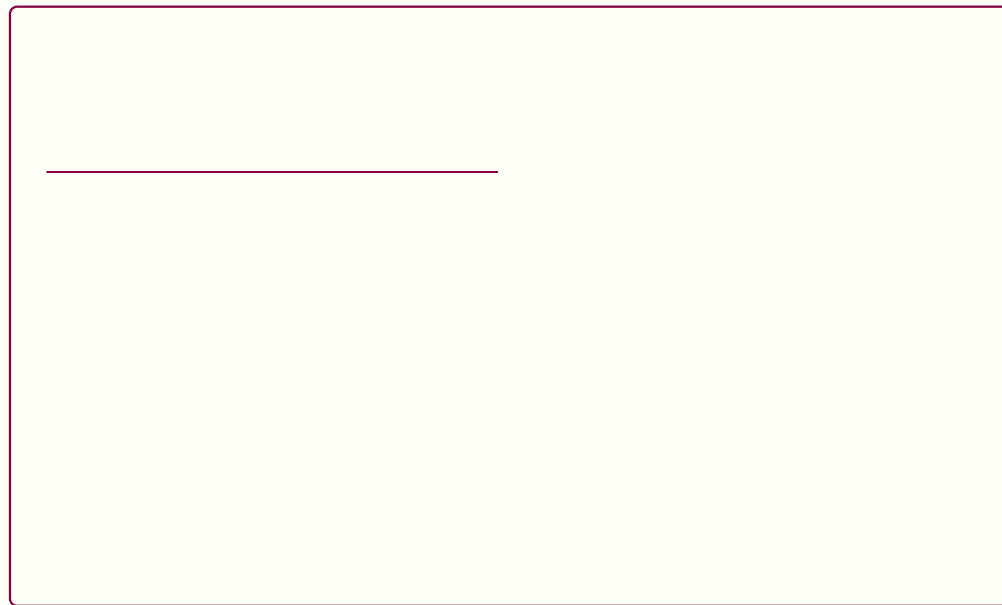


Title









Questions?

Back to your Projects

- Eins
- Zwei
- Polizei

Thank You

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July 2026

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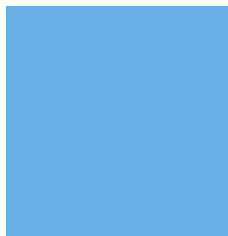
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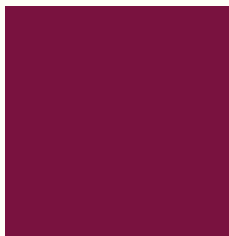
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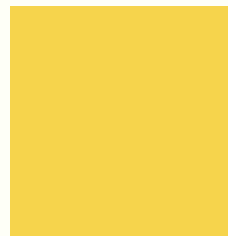
Dark purple
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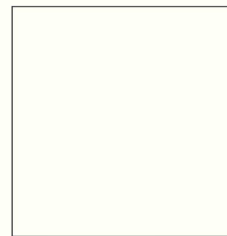
Light purple
#AE3156



Dark yellow
#EBA547



Light yellow
#F6D54D



Eggshell
#FEFF7